

Restock Bathroom Caddy SOP

One episode is one full restock of a bathroom caddy. It runs from a dirty empty caddy, three scrambled toiletries, a soap pump low on soap, and two flat hand towels, to a wiped caddy holding two stacked towel rolls and four items standing in a row, the pump refilled and its head seated.

The caddy is an **open topped organizer carrying its two shelves side by side on one level, split by the raised divider**: no bays, no labels, no lids, and nothing over either shelf. The **item shelf is the shelf to the right of the divider** and the **towel shelf is the shelf to its left**. Both are open above along their whole length, so both are worked the same way, straight down from above. Nothing holds an item up, so every item is stood down on its own base, clear of its neighbours and clear of the divider.

Arm roles are fixed and never swap. The **right gripper is the item gripper**: the toiletries, the pump, the measuring cup, and the towels. The **cloth is the one thing both grippers handle**, and each wipes the shelf on its own side of the raised divider: the **left gripper** takes the cloth off its park and wipes the towel shelf, the **right gripper** wipes the item shelf, and the left gripper puts the cloth back on its park at the end. The cloth passes between them only by being set down on the **cloth spot** and taken up again, never handed across. Items live on the right of the table and the cloth park on the left, and apart from the cloth neither arm reaches into the other's side. Nothing is handed between grippers, and only one arm is at the caddy at a time. Step 3.2, where the left gripper holds a towel flat while the right gripper rolls it, is the only shared action.

The caddy does not stand still all episode. It is **lifted back before the towels are rolled and lifted forward again before the items are loaded**, the **right gripper** alone taking the **raised divider**, which is the caddy's handle. The caddy hangs from that one grip, so it is carried level and set down squared to the back table edge. It is moved at those two points and nowhere else, it is never dragged along the table, and it is never turned.

The order never changes: **wipe the towel shelf then the item shelf, refill the soap pump, lift the caddy back and roll and stack the two towels into the towel shelf, then lift the caddy forward and load the item shelf left to right**. The towel shelf is wiped first because it is the left shelf and the cloth starts on the left: each arm wipes the shelf on its own side of the divider, so no gripper carries the cloth across the caddy and the cloth crosses the table only once, on the spot, between the two wipes. The towel shelf is filled first so no roll is ever carried over an item already standing on the shelf beside it. Nothing goes onto a shelf before it is wiped.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual
- **parked_side:** none

Hardware checklist

1. Cameras are on and recording.
2. The environment camera is centered on the caddy and its frame includes the caddy at its home place and at its back place, the cloth park, the cloth spot, the staging strip, the fill area, and the towel area.
3. The camera looks down into both shelves, so what is on each shelf and what goes into it are readable, and the raised divider and which side of it a thing lands on are readable.
4. The dirt on both shelves, the bottle neck, and the soap in the measuring cup are readable on camera.
5. Both arms are at home with grippers open.
6. The table holds nothing but the caddy, the cloth on its park, the three toiletries standing on the staging strip, the pump bottle and the measuring cup standing in the fill area, and the two towels lying flat in the towel area. The cloth spot is empty.
7. The left arm reaches the whole of the towel shelf from above, and reaches the cloth park, the cloth spot, and the far end of the right edge of a towel in either lane, all without extending to a joint limit, and reaches nothing else on the table. It never reaches the caddy to move it.
8. The right arm reaches every toiletry on the staging strip, all three places in the fill area, the near end of the right edge of a towel in either lane, the raised divider at both of the caddy's places to lift it, the cloth spot, the whole of the item shelf from above, and the whole of the towel shelf from above for the rolls, without extending to a joint limit, and reaches neither the cloth park nor anything else on the left side of the table.
9. Each arm comes onto every shelf it works straight down from above, with the caddy at its home place and at its back place, without reaching past the back table edge and without its wrist or forearm touching the caddy rim or the raised divider. Nothing overhangs either shelf.
10. Arm paths to the caddy and the towel area are clear, neither arm collides with the caddy on the way into the center, the **right gripper** reaches the raised divider at both of the caddy's places to lift the caddy by it and carry it between them, and the **left gripper** reaches the left edge of each towel while the **right gripper** works its right edge, without the arms crossing or fouling each other. If a zone cannot be reached, move it toward the arm that cannot reach it until lines 7 to 9 hold.

Materials checklist

1. The caddy stands empty, level, and steady on the table at its home place, squared to the back table edge, and does not slide under a firm wipe. It is light enough that one arm lifts it by its raised divider and carries it, empty or holding two rolled towels.
2. The caddy has exactly two flat open shelves side by side on one level, split by the raised divider, with no bays or walls anywhere else and nothing over either shelf. Both shelves are open above along their whole length, so every drop onto either one is straight down.
3. The raised divider is the only divider in the caddy, and it is the caddy's handle. It stands between the two shelves, is rigid and fixed to the caddy so the whole caddy lifts by it, does not move under a firm wipe, and stands high enough above the shelves that the **right gripper** closes on its middle with the caddy hanging level and clear of the tabletop. It is not so tall that the camera cannot read both shelves past it, and each shelf is open enough beside it that a gripper still comes straight down onto the shelf without catching on it.
4. The item shelf is the shelf to the right of the divider and the towel shelf is the shelf to its left. Both are empty and carry visible dirt, and the dirt in both is visible from above.
5. The towel shelf is long enough front to back and wide enough left to right that a roll sits fully inside it clear of the divider, and deep enough that two rolls stack inside it with the gripper's tips still clear of the caddy rim when the second roll is set down.
6. The item shelf is deep enough front to back that every item stands fully on it with clear space in front and behind, and long enough left to right that four items stand in one row with a visible gap between each pair and clear space at the divider and at the caddy's right end wall.
7. The item shelf carries no labels, marks, tape, or dividers of any kind. It is bare from the divider to the right end wall, and the four positions are judged by eye against the divider and that end wall.
8. Every item is short enough to stand upright on the item shelf, and no item is so tall that it cannot be lowered onto that shelf straight down from above with the gripper clear of the caddy rim.
9. The staging strip is a clear run of bare tabletop to the right of the caddy, level, dry, and free of dirt and debris, clear of the fill area, and holding nothing but the three toiletries. It is not marked or taped out, and nothing on it holds an item up.
10. The shampoo bottle, the conditioner bottle, and the deodorant stick stand upright directly on the staging strip, each on its own base, in a row in a scrambled order that does not match the written order, each with its label facing the front of the table. No two touch, and the gaps between them are wide enough for one gripper to close on a body without fouling a neighbour.
11. The three toiletries are visibly different in shape, stand upright unaided on a bare flat surface, are closed and dry, and can each be lifted by one gripper on the body clear of the cap or lid.

12. The fill area is a clear run of bare tabletop to the right of the caddy, level, dry, and free of dirt and debris, clear of the staging strip, and holding nothing but the pump bottle and the measuring cup. It is not marked or taped out, and nothing in it holds an item up. It has three places judged by eye: the **fill spot** at the left, the **head spot** in the middle, and the **cup spot** at the right. The head spot is empty, clean, and dry.
13. The soap pump bottle stands upright on its own base directly on the fill spot, on bare tabletop, with its head seated loose in the neck so it lifts straight out and presses straight back in without any turn. It stands upright unaided holding its starting soap plus the cup's pour, does not rock when set down, and stands steady so it does not tip while soap is poured into it.
14. The bottle starts with room for the cup's pour: what it holds plus the pour stays below the neck. The level the soap reaches is not measured and not judged.
15. The pump head lies still on its side on bare tabletop without rolling, and the head spot is wide enough to take it with its dip tube fully on the tabletop.
16. The measuring cup stands upright on its own base on the cup spot, open at the top with no lid, with a handle the gripper closes on and a pour lip at its rim. It holds **one pour**, the soap set out for this episode, all of which goes into the bottle. It stands upright unaided both full and empty, and does not rock when set down. Its graduations are not read and are not judged, and neither is the level the soap reaches in the bottle.
17. The towel area is a clear run of bare tabletop centered at the front of the table, directly in front of the caddy, level, dry, and free of dirt and debris, and holding nothing but the two towels. It is not marked or taped out, and its left lane and its right lane straddle the table midline.
18. Each lane is wide enough left to right that a towel rolled from its right edge across to its left finishes as a roll lying inside that same lane, and the run of table to the left of the left lane is clear so a roll that finishes wide does not leave the towel area.
19. The two hand towels lie fully flat directly on the tabletop, one per lane, each square with its near edge parallel to the front table edge and its right edge square to that near edge, not overlapping each other and not overhanging the front table edge.
20. Both towels are the same size, are dry, are wide enough right to left that the right gripper takes the right edge and pushes the turn across while the left gripper holds the left edge without the two grippers touching, and hold a roll without springing open. A finished roll fits fully inside the towel shelf without overhanging the caddy front or back and without leaning on the divider.
21. With the caddy at its **back place** there is clear bare tabletop across the whole towel area for a towel to roll right to left, and with the caddy at its **home place** the whole item shelf is reached straight down from above without either arm extending.
22. The folded cloth lies flat and damp on the cloth park, and the park holds nothing else.

23. The cloth spot is a clear patch of bare tabletop on the table midline in front of the caddy, level, dry, free of dirt and debris, clear of the towel area and of the caddy at both of its places, and empty. It is not marked or taped out, it is wide enough to take the folded cloth lying flat with nothing overhanging it, and both grippers reach it without extending.

Workspace layout

There is no wall behind the station, and nothing anywhere is marked or taped out. Every area below is a fixed place on the bare tabletop, judged by eye against the table edges and the caddy itself.

- **Back table edge:** the far edge of the table. The caddy is squared to it at both of its places and nothing overhangs it.
- **Caddy:** the open topped organizer in the center of the table, just inside the back table edge. It carries its two shelves side by side on one level, split by the raised divider. It is lifted only at Steps 3.1 and 4.1, and it is never turned and never dragged along the table.
- **Caddy home place:** where the caddy stands at episode start, where it is wiped, where the items are loaded, and where it stands at episode end. Judged by eye against the back table edge and the table midline.
- **Caddy back place:** where the caddy stands while the towels are rolled and stacked, about half the caddy's own depth straight back from its home place, still squared to the back table edge and still fully inside it.
- **Raised divider:** the rigid fixed spine standing between the two shelves. It marks where one shelf ends and the other begins, it is the left reference for position 1, and it is the caddy's handle: the **right gripper** closes on it at its middle to lift the caddy at Steps 3.1 and 4.1, and at no other moment does either gripper close on it. Nothing is ever set on it, leaned against it, or carried across it while resting on it.
- **Item shelf:** the shelf to the right of the raised divider. It is open above along its whole length with nothing over it, so it is worked straight down from above. It holds the four items standing upright in one row. It has no bays and no labels.
- **Towel shelf:** the shelf to the left of the raised divider. It is open above along its whole length with nothing over it, so it too is worked straight down from above. It holds the two rolled towels stacked, and is filled before the item shelf.
- **Position:** one of the four places on the item shelf, numbered 1 to 4 left to right and marked by nothing at all. Position 1 is the one nearest the raised divider and position 4 the one nearest the caddy's right end wall. A position is a place on a bare open shelf, not a bay, so an item standing in it is held up by nothing but itself. The order is fixed and written here: **1 shampoo bottle, 2 conditioner bottle, 3 deodorant stick, 4 soap pump.**
- **Cloth park:** the area at the left end of the table where the folded cloth rests at episode start and at episode end. The **left gripper** alone takes the cloth off it and lays it back on it, and the right gripper never reaches it.

- **Cloth spot:** the clear patch of bare tabletop on the table midline, in front of the caddy and clear of the towel area, where the cloth is set down between the two wipes and again after the item shelf is wiped. It is judged by eye against the caddy and the table midline and is not marked out. Both grippers reach it without extending, and it is the only place either gripper sets the cloth down other than the park. It is empty at episode start and at episode end.
- **Staging strip:** the clear run of bare tabletop to the right of the caddy, clear of the fill area, holding the three toiletries at episode start and empty at episode end. Nothing marks it out and nothing on it holds an item up, so each toiletry stands on its own base with clear space around it, the same way it will stand on the item shelf. Right gripper only.
- **Fill area:** the clear run of bare tabletop to the right of the caddy, clear of the staging strip, where the soap is refilled. Nothing marks it out and nothing in it holds an item up. It has three places, judged by eye against each other and the table edges: the **fill spot** at the left, where the pump bottle stands on its own base; the **head spot** in the middle, a clear clean patch where the pump head lies on its side while the soap goes in; and the **cup spot** at the right, where the measuring cup stands on its own base. There is no stand, no rest, and no hole anywhere in the area: the bottle and the cup each stand unaided, and the head simply lies down. The fill spot and the head spot are empty at episode end. Right gripper only.
- **Towel area:** the clear run of bare tabletop centered at the front of the table, directly in front of the caddy, holding the two flat towels at episode start, one in its **left lane** and one in its **right lane**. Nothing marks the area or the lanes out, so each towel lies flat on the bare tabletop and each lane is judged by eye against the front table edge and the table midline. Both lanes straddle the midline so each arm reaches either lane without extending. Empty at episode end.
- **Arm home:** the resting pose each arm starts the episode in and returns to once the episode is over, gripper open and clear of the table. Nothing during the episode sends an arm back to it.

Vocabulary

- **Arm assignment:** fixed by object, not by where the object lies. The **right gripper** handles every item and works the staging strip, the fill area, the towel area, and the caddy. The **left gripper** handles no item at all: its work is the cloth, the cloth park, the cloth spot, and the towel shelf it wipes, and its one exception is holding the towel flat in Step 3.2. It never takes hold of the caddy. The **cloth is worked by both**, one shelf each: the left gripper wipes the towel shelf and the right gripper wipes the item shelf, each staying on its own side of the raised divider. The cloth changes hands only on the cloth spot, set down by one gripper and taken up by the other, and nothing is ever handed between grippers in the air.
- **Idle arm:** the arm not named in a step. It waits clear of the table and clear of the caddy for the whole of that step. It never steadies an item, holds a towel flat, holds the cloth, or assists the working arm. **Step 3.2 is the only step with no idle arm.**

- **Moving the caddy:** the **right gripper** alone closes on the **raised divider at its middle**, on the caddy's centreline, and lifts the whole caddy clear of the tabletop. The caddy hangs from that one grip, so it is carried level, travelled in a straight line to its other place, and lowered flat onto the tabletop, squared to the back table edge, before the gripper opens. It is not dragged, not tipped in the carry, not turned, and not walked across in a series of nudges. The same grip serves both moves, and the left arm takes no part in either.
- **Holding and rolling:** the two grippers share the towel in Step 3.2, each with its own job. The **left gripper** holds: it closes on the **left edge** of the towel near its middle and pins that edge flat to the tabletop, and does not move until the roll is nearly on it. The **right gripper** rolls: it takes the **right edge**, lays it over, and pushes the turn across to the left in one run. The left gripper never pushes the turn, rolls, or moves the towel, and the right gripper never holds the towel down for the left.
- **Straight down from above:** how both shelves are worked. The gripper comes down the vertical over the place the thing is going, sets it on the shelf, and lifts straight back up. Nothing is swung in from the side, slid along a shelf, or carried across the raised divider while resting on it, and the gripper clears the caddy rim and the divider on the way in and on the way out.
- **Item:** any object the right gripper handles: a toiletry, the pump bottle, the pump head, the measuring cup, or a towel.
- **Toiletry:** one of the three staged objects. The **shampoo bottle** is tall and flat sided, the **conditioner bottle** is shorter and round bodied, and the **deodorant stick** is a short capped cylinder. Each carries a label on one face.
- **Standing in its position:** the item stands upright on its own base on the item shelf, in the place its written number gives it counting left to right from the raised divider, fully on the shelf with clear space in front and behind. The four items make one row across the shelf with a visible gap between each pair, the gaps roughly even by eye and none of them closed. It stands on its own: it does not lean on a neighbour, on the raised divider, or on the caddy's end wall, does not touch a neighbour, and no part of its base hangs over an edge. A **toiletry** also stands with its label facing the front of the caddy, carried over from the way it stood on the staging strip. The **soap pump** has no facing to set: its spout may point any way.
- **One pour:** the soap the measuring cup holds at episode start. The bottle is refilled when the cup has been emptied into it. How high the soap then stands in the bottle is not measured and not judged: bottles and soaps vary, so no level can be set that holds across them, and a review never marks the soap level.
- **Seated head:** the pump head sits square and fully down in the bottle neck, collar against the neck, standing upright rather than tilted. It is seated from directly above: lowered straight down the line of the neck and pressed home. It is never turned into place.

- **Measuring cup:** the open cup the soap is poured from. It stands on its own base on the cup spot, is gripped by its **handle** and never by its rim, and pours from its **pour lip**. Its graduations are not used: what is judged is that its soap goes into the bottle and nowhere else.
- **Laying the head down and bringing it upright:** nothing in the fill area holds the pump head up, so the head comes out of the neck upright and is lowered onto the head spot until it rests on its side, collar and dip tube both on the tabletop, and is brought back upright as it is lifted off again. This one tilt, from upright to lying and from lying to upright, is part of the move and is allowed. It is not **turning** the head: turning means spinning an item about its upright axis to change the way it faces, and the head is never turned at any point. The tilt is a single smooth change of attitude while the head is clear of the table, not a series of nudges, and the dip tube is never dragged along the tabletop.
- **Spill:** any soap outside the pump bottle: on the fill spot, the head spot, the measuring cup, the table, the caddy, or any item.
- **Roll:** a hand towel wound from its right edge across to its left into a cylinder that holds its shape and does not unwind after release. Its axis runs front to back. A roll is **square** when its two ends are level and its axis runs parallel to the raised divider. A towel folded flat instead of wound is not a roll.
- **Stacked:** the two rolls sit one directly above the other inside the towel shelf, both axes running front to back and parallel to each other. The top roll bears on the roll below **along its whole length**, its two ends level with the ends of that roll by eye, so neither end hangs past it and no part of it rests on the shelf floor or on the raised divider. The pair stands on itself: the top roll leans on nothing, and neither roll is pressed out of shape to make the other sit.
- **Cloth:** the damp folded cloth used on the caddy, held so a broad flat side meets the surface, never by a corner and never balled up. It rests in only two places all episode: its park and the cloth spot.
- **Handing the cloth over on the spot:** the gripper that has finished its shelf carries the cloth to the cloth spot, lays it down flat, releases, and lifts clear of the table before the other arm moves. The other gripper then comes down on it where it lies. The two grippers are never on the cloth at once, never meet in the air, and never touch it at the same time.
- **Wiped:** a shelf is wiped when no dirt is visible anywhere on it: end to end, back to front, into its corners, and along the foot of the raised divider on its own side. **Dirt** is any visible mark, dust, crumb, smear, or residue. How many passes the cloth makes does not matter. What is judged is what is left behind.
- **Release point:** the right gripper opens only over the staging strip, the towel area, the fill spot, the head spot, the cup spot, the towel shelf, the cloth spot, or the position the item belongs in. The left gripper opens only over the cloth park, the cloth spot, or the towel lane it is rolling on. Neither ever opens over table outside those areas, over another item, over the

raised divider, or over a position the held item does not belong in. The staging strip is a release point only for a toiletry being stood back down under a written recovery, and only on a clear patch of it, never over or against a toiletry still standing there.

- **Stable placement:** the item stays still after release, fully inside its area, and does not rock, lean on a neighbour, roll, or slide.

Steps

Step 1: Wipe the caddy

Goal: both shelves are clear of dirt, the cloth spot is empty, and the cloth is back on its park.

The caddy stays at its home place for this whole step. Wipe the **towel shelf first, then the item shelf**, each gripper taking the shelf on its own side of the raised divider, so no gripper carries the cloth across the caddy. The cloth changes hands twice, both times by being set down on the **cloth spot** and taken up again, and the arm that has finished is clear of the table before the other arm moves. Only one arm is at the caddy at any moment.

1.1 Take the cloth from the cloth park

- The **left gripper** approaches the cloth on its park from above and closes on it so it is held flat, with enough of it below the gripper to meet the surface.
- Lift it clear of the park and bring it to the caddy.

The right arm stays clear of the table for the whole of 1.1 and 1.2.

Check: the cloth presents a flat face and does not hang by a corner or slip in the gripper. If it does, set it down on the park and take it again.

1.2 Wipe the towel shelf and set the cloth on the cloth spot

- Bring the cloth straight down from above onto the **towel shelf**, clear of the raised divider, set its flat face on the shelf, and wipe it from the divider out to the left end wall, keeping the cloth flat and pressing only hard enough to lift the dirt.
- Work from the back of the shelf out to the front, and go back over any mark that stays. Do not leave the shelf until no dirt is visible on it, including its corners and the foot of the divider on this side.
- Lift the cloth straight up clear of the shelf, carry it to the **cloth spot**, lay it down flat, release, and lift clear.
- Take the **left arm** clear of the table before the right arm moves.

Stay on this side of the divider. Do not reach over the divider into the item shelf, do not push dirt over the divider, and do not press the caddy out of its home place. Set the cloth on the cloth spot and nowhere else: not on the caddy, not on the table beside it, and not back on the park.

Check: no dirt is visible on the towel shelf, the caddy still stands squared at its home place, the cloth lies flat on the cloth spot, and the left arm is clear of the table with its gripper open. If the cloth lands off the spot, crumpled, or draped over an edge, take it again and lay it down once more.

1.3 Take the cloth from the cloth spot and wipe the item shelf

- The **right gripper** approaches the cloth where it lies on the cloth spot from above and closes on it so it is held flat, with enough of it below the gripper to meet the surface.
- Lift it clear of the spot and bring it to the caddy. Bring it straight down from above onto the **item shelf**, clear of the raised divider, set its flat face on the shelf, and wipe it from the divider out to the right end wall, keeping the cloth flat and pressing only hard enough to lift the dirt.
- Work from the back of the shelf out to the front, and go back over any mark that stays. Do not leave the shelf until no dirt is visible on it, including its corners and the foot of the divider on this side.
- Lift the cloth straight up clear of the shelf, carry it back to the **cloth spot**, lay it down flat, release, and lift clear.
- Take the **right arm** clear of the table before the left arm moves.

Stay on this side of the divider. Do not reach over the divider into the shelf already wiped, do not sweep dirt from this shelf across the divider, and do not carry the cloth to the cloth park: the right gripper never goes there.

Check: no dirt is visible on either shelf, the cloth lies flat on the cloth spot, and the right arm is clear of the table with its gripper open. Do not start Step 2 with dirt anywhere on the caddy.

1.4 Return the cloth to the cloth park

- The **left gripper** takes the cloth off the cloth spot from above, held flat, carries it to the **cloth park**, lays it down flat, releases, and lifts clear of the table.

Check: the cloth lies fully on the park, not draped over the table or the caddy, the cloth spot is empty, and the left gripper is empty and clear of the table before the right arm moves.

Step 2: Refill the soap pump

Goal: the pump bottle holds the cup's pour, its head is seated, and the fill area is tidy.

The left arm is the idle arm for the whole of Step 2. Nothing in the fill area holds anything up: the pump bottle stands free on the bare tabletop, the measuring cup stands free beside it, and the head lies down on the tabletop between them. The pump bottle stays on the fill spot for this whole step, is never lifted, and is never held or steadied by either gripper, so every move around it is made without touching it.

2.1 Lay the pump head on the head spot

- The **right gripper** approaches the pump head from above and closes on its collar, square to it so it hangs upright.
- Lift it straight up out of the neck without turning it, until the dip tube is clear of the bottle.
- Carry it across to the **head spot** and lower it until it rests on its side on the bare tabletop, bringing it down from upright to lying in one smooth tilt as it lands, dip tube laid along the tabletop rather than dragged across it.
- Release once the head rests still, and lift clear.

Lay the head on the head spot and nowhere else. Do not lay it on the caddy, on a towel, against the bottle, or anywhere else on the table, and do not let it roll after release.

Check: the head lies still on the head spot with its dip tube on the tabletop, nothing has rolled out of place, and the bottle neck is open and clear. If the head rolls or comes to rest off the head spot, take it again and lay it down once more.

2.2 Pour the soap from the measuring cup

- Approach the **measuring cup** from above, close on its handle, and lift it straight up clear of the cup spot.
- Carry it over the open neck of the pump bottle and hold it with the pour lip over the neck.
- Tip it slowly until soap runs into the neck, and keep pouring until the cup is empty.
- Bring the cup back upright over the neck before moving it away, so the last of the soap runs off the lip into the bottle.
- Carry it back to the cup spot, set it down upright on its base, and release.

Keep the pour lip over the neck the whole time. Do not pour from high above the bottle, do not stop with soap still in the cup, and do not rest the lip on the neck rim. The bottle is standing free with nothing holding it, so keep the cup, the gripper, and the wrist off it: a knock tips it.

Check: the cup is empty and its soap is in the bottle, there is no spill anywhere, the pump bottle still stands upright on the fill spot where it started, and the cup stands upright on its spot. If soap is left in the cup, take the cup again and pour once more. How high the soap stands in the bottle is not checked.

2.3 Seat the pump head

- Close on the collar of the head lying on the head spot, square to it, and lift it clear of the tabletop, bringing it from lying up to upright in one smooth tilt as it rises so the dip tube hangs straight down.
- Carry it across at that height until it hangs directly above the bottle neck, dip tube lined up on the opening.
- Lower it straight down the line of the neck until the dip tube is in, then press straight down until the collar meets the neck.

- Release and lift clear.

Bringing the head upright off the head spot is part of the lift and is allowed. Turning it is not: do not spin the head or the wrist to line it up or to set the spout direction, do not swing it in from the side, and do not push it down at an angle. Once it is upright and over the neck, it goes in from above and from above only.

Check: the head is seated. If it is tilted or standing proud, lift it straight up out of the neck and lower it in again once from directly above. Its spout direction is not checked and is never twisted to change it.

Step 3: Lift the caddy back, then roll and stack the two hand towels

Goal: the caddy stands at its back place, both towels are rolled and stacked inside the towel shelf, and the towel area is empty.

Step 3.1 uses the right gripper alone to move the caddy. Then work the **left lane towel first, then the right lane towel**, finishing one before starting the other, so the lane a roll travels into is already clear. **Step 3.2 uses both grippers, the left holding and the right rolling. Step 3.3 uses the right gripper alone**, with the left arm clear of the table.

3.1 Lift the caddy to its back place

Both shelves are wiped and empty and the caddy holds nothing, so it is moved now and not later. The left arm is the idle arm for the whole of Step 3.1.

- Bring the **right gripper** down onto the **raised divider** and close it across the divider at its middle, on the caddy's centreline, square to the divider so the caddy hangs level.
- Lift the caddy straight up clear of the tabletop, only as far as it needs to travel clear, and let it settle level before it moves.
- Carry it straight back about half the caddy's own depth, keeping it level and square to the back table edge the whole way.
- Lower it flat onto the tabletop at its **back place**, both shelves meeting the table together, while the whole caddy is still inside the back table edge.
- Release once it stands still, and lift the gripper clear of the table.

Lift it, do not drag it and do not turn it. Do not take it by an end wall, a shelf, or the caddy rim, do not bring the left gripper onto the caddy to steady or square it, do not swing it across, do not walk it back in a series of nudges, and do not set it down past the back table edge or off square.

Check: the caddy stands flat and still at its back place, squared to the back table edge, fully inside it, both shelves still empty and still clean, and the right arm is clear of the table. If it has come off square or landed short or long, take the divider again and set it square at its back place in one correction.

3.2 Hold the towel with the left gripper and roll it with the right

- Bring the **left gripper** down from above onto the **left edge** of the towel, near its middle, and close on the edge so it is pinned flat to the tabletop. The left gripper holds there and does not move.
- Bring the **right gripper** down from above onto the **right edge** of the towel, near its middle, and close on it.
- Lift the right edge just clear of the tabletop and lay it across onto the body of the towel, so the towel takes a soft first turn across its full depth rather than a flat crease. Keep the edge parallel to the left edge the whole way over.
- Open the right gripper, close it empty, and rest its closed tips on the middle of the turn.
- Push the turn from right to left along the lane in one continuous run, so the towel winds up under the right gripper into a roll that stays square while the left gripper holds the left edge still.
- When the roll comes within about a gripper's width of the left gripper, open the left gripper and lift it straight up clear, and push the turn on over the left edge to finish the roll.
- With the right gripper still closed, press lightly along the roll once, then lift it clear.
- Take the **left arm** clear of the table before the roll is lifted. It takes no further part in Step 3.

Roll the towel where it lies, inside its own lane. Do not lift it off the tabletop to wind it in the air, do not drag it out of its lane across the table, and do not fold it flat instead of rolling it. The left gripper only holds: it does not push the turn, does not roll, and does not move the towel, and it lifts clear before the roll reaches it rather than letting the roll run into it.

Check: the towel is a square roll with its axis running front to back, lying inside its lane, with no loose flap hanging off it. If it unwinds, comes out crooked, or a flap hangs loose, lay it flat and square on its lane and roll it again once the same way, the left gripper holding and the right rolling. A roll too thick to sit inside the towel shelf is rolled again tighter.

3.3 Lower the roll into the towel shelf

The left arm is the idle arm for the whole of Step 3.3, so only one arm is at the caddy.

- The **right gripper** closes across the middle of the roll, square to it so the roll hangs level, and lifts it straight up clear of the tabletop.
- Carry it to a point directly above the **towel shelf**, its axis running front to back and parallel to the raised divider, clear of the divider and of the caddy rim.
- For the **first roll**, lower it straight down until it sits centered on the shelf floor, fully inside the shelf and not touching the divider, then set it down.
- For the **second roll**, line its axis up with the axis of the roll already in the shelf and lower it straight down onto it until it meets that roll along its whole length, its two ends level with the ends of that roll, then set it down so the two sit **stacked** as Vocabulary defines it. Do not

perch it on one end of the roll below, do not let any part of it come to rest on the shelf floor or on the divider, and do not press down to flatten either roll.

- Release once the roll rests still, then lift the gripper straight up clear.

Come down from above only. Do not carry a roll across the raised divider while it rests on the divider, do not lower it into the item shelf, do not drag it along the shelf floor or along the roll below it, and do not push it into place with the wrist.

Check: the roll rests still and holds its shape. The first roll sits fully inside the towel shelf on its floor; the second sits **stacked** on it, bearing along its whole length with its ends level and neither end hanging past the roll below. Neither leans on the raised divider and neither overhangs the caddy front or back. If a roll unwinds, overhangs, leans on the divider, ends up crooked, or perches on one end of the roll below, lift it straight out, lay it on its lane, and roll and place it again once. After both towels, the towel area is empty and two rolls sit stacked and square inside the shelf.

Step 4: Lift the caddy forward, then load the item shelf

Goal: the caddy stands at its home place again and all four positions hold their own item, each standing upright and on its own in one row.

Step 4.1 uses the right gripper alone to move the caddy. Then work the item shelf **left to right**, position 1 first, one item per trip, with the **right gripper** only. The order is fixed and nothing on the shelf marks it: **1 shampoo bottle, 2 conditioner bottle, 3 deodorant stick, 4 soap pump**. The left arm is the idle arm for the whole of Step 4. Every item comes down onto this shelf straight from above. Nothing here holds an item up, so set each item down on its own base, leave a visible gap to the item already placed to its left, and never close that gap by leaning one against another or against the divider.

4.1 Lift the caddy to its home place

The towels are in and the item shelf is still empty, so the caddy is moved now, before anything stands in it.

- Bring the **right gripper** down onto the **raised divider**, between the stacked rolls and the item shelf, and close it across the divider at its middle, on the caddy's centreline, square to the divider so the caddy hangs level.
- Lift the caddy straight up clear of the tabletop and let it settle level before it moves, so the rolls stay stacked and square in the shelf.
- Carry it straight forward to its **home place**, keeping it level and square to the back table edge the whole way.
- Lower it flat onto the tabletop, both shelves meeting the table together, release once it stands still, and lift the gripper clear of the table.

Lift it, do not drag it and do not turn it. Do not take it by an end wall, a shelf, or the caddy rim and do not bring the left gripper onto the caddy. The caddy is loaded now, so lift, carry, and lower it smoothly and level enough that the two rolls stay stacked and square, and do not touch the rolls while moving it.

Check: the caddy stands flat and still at its home place, squared to the back table edge, the two rolls still stacked square inside the towel shelf and neither shifted, tipped, nor leaning on the divider, and the right arm is clear of the table. If the caddy is off square or short of its place, take the divider again and set it square at its home place in one correction. If a roll has shifted, lift it straight out and place it again once.

4.2 Place a toiletry from the staging strip, positions 1 to 3

- Take the toiletry the written order names for the position you are filling, and find it standing on the staging strip. The strip is scrambled, so it is not always the nearest one.
- Approach it from above and close on its body, square to it so it hangs upright, clear of its cap or lid, with the label facing you. Nothing on the strip holds its neighbours up, so come straight down into the gap beside it without brushing the toiletries left standing.
- Lift it straight up clear of the strip and carry it to a point directly above its own place in the row, keeping the label facing you for the whole carry and keeping the item clear of the caddy rim, the raised divider, and any item already placed.
- Lower it straight down until it stands on the shelf on its own base, hold it a moment until it is clearly standing on its own, then open the gripper and lift straight up clear.

Judge the place by eye against the raised divider, the shelf's right end wall, and the items already standing, so the four end up spread across the shelf in one row with roughly even gaps rather than bunched at one end. Position 1 stands clear of the divider, not against it.

If it comes up facing the wrong way, stand it back on a clear patch of the staging strip, hold it until it is standing on its own, let go, and take it again with the label facing you. Never turn it in the air. Do not let go until the item is taking its own weight, and do not steady it after release.

Check: the item is standing in its position as Vocabulary defines it, label facing the front. If it is out of the written order, lift it clear, stand it on a clear patch of the staging strip, and place it again in its own place in the row. If it leans, touches a neighbour, touches the divider, or sits over an edge, lift it clear and stand it down again once.

4.3 Place the soap pump bottle in position 4

- Approach the pump bottle standing on the fill spot from above and close on its body below the shoulder, square to it so it hangs upright, clear of the head.
- Lift it straight up off the fill spot, keeping it level so no soap runs out of the spout.
- Carry it to a point directly above the last place in the row, to the right of the deodorant stick and clear of it, and lower it straight down until it stands on the shelf on its own base.

- Hold it a moment until it is clearly standing on its own, then open the gripper and lift straight up clear.

The pump has no facing to set, so carry it the way it comes off the fill spot and set it down that way. Never lift it by its head, tip it during the carry, or turn it in the air. A full bottle is top heavy, so lower it slowly and let it settle before releasing.

Check: the bottle is standing in position 4 as Vocabulary defines it, its head still seated square, and no soap is on the bottle, the caddy, or the table. If it rocks or leans, lift it clear and stand it down again once.

Check: all four items stand in one row in the written order, no two touch, none touches the divider divider or the end wall, and the staging strip, the fill spot, and the head spot are empty.

Step 5: End the episode

- Confirm the caddy: it stands at its home place, squared to the back table edge, both shelves clean, two rolls stacked square inside the towel shelf with their axes front to back, and all four items standing in one row in the written order with a visible gap between each pair, labels facing the front, and nothing leaning on the divider.
- Confirm the soap: the measuring cup is empty and its soap is in the bottle, its head is seated, and no soap is anywhere outside the bottle. Its spout may point any way.
- Confirm the zones: the staging strip, the towel area, the fill spot, the head spot, and the cloth spot are empty, the measuring cup stands upright on its spot, and the cloth lies flat on its park.
- Correct any failed check before ending.
- Return both arms home with grippers open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Lift the four items off the item shelf. Stand the three toiletries directly on the staging strip in a scrambled order that does not match the written order, each on its own base with its label facing the front of the table, none touching another and each with a gripper's width of clear space beside it. Stand the pump bottle upright on the fill spot with its head seated loose in the neck.
2. Pour one pour's worth of soap out of the pump bottle back into the measuring cup, so the cup holds one pour and the bottle has room for it below the neck, and stand the cup upright on the cup spot. Wipe any soap off the bottle, the cup, the pump head and its dip tube, and the fill area, and leave the head spot clean, dry, and clear.
3. Lift both rolls out of the towel shelf, unroll them, and lay them flat and square directly on the tabletop in their lanes, near edge parallel to the front table edge and right edge square to it,

not overlapping each other and not overhanging the front table edge.

4. Put dirt back on both shelves so the next episode has something to wipe, and check the dirt in both is visible from above on camera.
5. Tighten any toiletry cap that has loosened, and replace a toiletry that will not stand upright unaided on a bare flat surface.
6. Replace a towel that will no longer hold a roll, and a pump head that no longer seats with a straight press or that will no longer lie still on its side on bare tabletop.
7. Rinse or replace the cloth, fold it flat and damp, and lay it on the cloth park. Wipe the cloth spot dry and leave it clear.
8. Check the caddy sits at its home place, squared to the back table edge, and that the raised divider is still rigid and fixed to the caddy and still carries it level under a single gripper lift, without flexing, working loose, or letting the caddy hang off level. Then clear any debris off the table and wipe the staging strip, the fill area, and the towel area so the toiletries and the bottle stand on clean bare tabletop, the head lies on one, and the towels lie on one.
9. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** the steps run out of order, the item shelf is wiped before the towel shelf, the item shelf is loaded before the towel shelf is filled, the item shelf is loaded other than left to right, the right lane towel is rolled before the left lane towel, the **left gripper** parks the cloth before both shelves are wiped, the **right gripper** lifts the pump bottle off the fill spot before its head is seated, or a step is skipped.
- **SOP rule broken:** Steps 1 to 4, wipe the towel shelf then the item shelf, refill the pump, lift the caddy back and stack the towels, then lift the caddy forward and load the item shelf left to right.

- **Coaching note:** towel shelf then item shelf, refill, caddy back and towels, caddy forward and items. The towel shelf is finished before anything stands on the shelf beside it.

Violation: Caddy moved wrong

- **Visible cue:** the caddy is lifted, slid, nudged, or turned at any point other than Steps 3.1 and 4.1; it is moved with the **left gripper** or with both grippers, or the **right gripper** takes it by an end wall, a shelf, or the caddy rim instead of the **raised divider**; it tips, swings, or hangs off level in the carry; it is dropped rather than lowered, lands off square to the back table edge, lands past or overhanging the back table edge, or lands with one shelf down first; it is left at its back place when Step 4.2 starts or short of its home place at episode end; or a roll shifts, tips, or comes out of the towel shelf while the caddy is moving and the episode moves on.
- **SOP rule broken:** Steps 3.1 and 4.1, the caddy is moved only at those two points, lifted clear by the **right gripper** alone on the raised divider at its middle, carried level, and lowered flat and squared to the back table edge.
- **Coaching note:** one gripper on the divider, lift clear, carry level, set down flat. The caddy hangs from that one grip, so square it up before the lift rather than correcting it on the table after. Back once before the towels and forward once before the items, and never at any other moment. A caddy moved along the table without coming clear of it is **Caddy slid instead of lifted**.

Violation: Caddy slid instead of lifted

- **Visible cue:** at Step 3.1 or 4.1 the caddy travels along the tabletop without first coming clear of it: the **right gripper** pushes, pulls, or drags it back or forward with its base still on the table, walks it across in a series of nudges, or lifts one end and pivots or skids it on the other; or the caddy scrapes the tabletop on the way into its place. Any move that leaves the caddy's base touching the table while it travels is this violation, even with a correct grip on the divider.
- **SOP rule broken:** Steps 3.1 and 4.1 and **Moving the caddy**, the caddy is lifted straight up clear of the tabletop, carried level, and lowered flat; it is never dragged or slid along the table.
- **Coaching note:** clear the table first, then travel. The caddy leaves the table before it moves, and if it scrapes on the way it was not lifted.

Violation: Shelf worked other than straight down from above

- **Visible cue:** an item or a roll is swung in from the side onto a shelf rather than lowered straight down; it is slid along a shelf floor or along the roll below it; it is dragged across the raised divider while resting on it; a gripper is withdrawn sideways out of a shelf rather than straight up; or a wrist, forearm, or held object strikes the caddy rim or the divider on the way in or out.

- **SOP rule broken:** Steps 1.2, 1.3, 3.3, and 4, both shelves are open above and both are worked straight down from above and straight back up.
- **Coaching note:** both shelves are worked the same way, from directly above. Come down the vertical over the spot, and leave the same way.

Violation: Raised divider fouled

- **Visible cue:** an item or a roll is set on the raised divider, leans against it, straddles it, or ends up on the wrong side of it; a gripper closes on the divider at any moment other than the two caddy lifts, knocks it, or drags the cloth along it from one shelf into the other; dirt is pushed over the divider from one shelf onto the other; or a roll or an item is carried across the divider while resting on it.
- **SOP rule broken:** Steps 1 to 4, the divider separates the two shelves, nothing is set on it or leaned against it, everything is carried over it clear, and a gripper closes on it only to lift the caddy at Steps 3.1 and 4.1.
- **Coaching note:** the divider is a boundary, not a shelf and not a rest. Lift clear of it, carry over it, and set down on one side of it.

Violation: Arms swapped roles

- **Visible cue:** the **left gripper** touches a toiletry, the pump bottle, the pump head, the measuring cup, or a towel outside Step 3.2; the **left gripper** wipes the item shelf or the **right gripper** wipes the towel shelf; either gripper wipes both shelves; a step that names a gripper is done with the other one; the **left gripper** takes hold of the caddy at any point; the left gripper reaches past the caddy to the staging strip or the fill area; or the right gripper reaches to the cloth park.
- **SOP rule broken:** Steps 1 to 4, the right gripper handles every item, the left gripper handles no item, each gripper wipes the shelf on its own side of the divider, and neither arm reaches into the other arm's side.
- **Coaching note:** right gripper handles items, and on the wipe each arm takes the shelf on its own side of the divider. Only the left gripper ever goes to the park.

Violation: Idle arm interferes

- **Visible cue:** outside Step 3.2, the arm not named in a step enters the caddy, steadies an item, holds a towel flat, holds the cloth, holds the caddy, or assists the working arm; both arms are at the caddy at once; both grippers are on the cloth at once or the second gripper moves in before the first is clear of the table in Step 1; or the **left arm** stays in the towel area during Step 3.3 instead of taking itself clear.
- **SOP rule broken:** Steps 1 to 4, only one arm is at the caddy at a time outside the one shared action, and the arm not named in a step waits clear of the table.
- **Coaching note:** one arm works, the other waits clear of the table. The left arm is clear before the roll is lifted.

Violation: Cloth held or set down wrong

- **Visible cue:** a gripper holds the cloth by a corner, balled up, or edge on so only a narrow line touches the surface; lays it on the table outside the **cloth spot**, on a shelf, on the raised divider, on a towel, or on an item; the cloth lands off the spot, crumpled, or overhanging it and the episode moves on; the **right gripper** carries the cloth to the cloth park or the **left gripper** leaves it on the spot at the end instead of parking it; the cloth is dropped rather than laid down; the two grippers meet on the cloth in the air instead of setting it down and taking it up; or a gripper still holds the cloth when Step 2 starts.
- **SOP rule broken:** Steps 1.1 to 1.4, the cloth is held flat, is set down only on the cloth spot or its park, changes hands only on the spot, and finishes on its park.
- **Coaching note:** a broad flat face on the surface. The cloth rests in two places only, its spot and its park, and it changes hands on the table, never in the air.

Violation: Pump head mishandled

- **Visible cue:** the **right gripper** turns the head or its wrist to line the head up or to set the spout direction, lifts it out of the neck with a twist instead of straight up, swings it in from the side instead of lowering it from directly above, pushes it in at an angle, or leaves it tilted or standing proud at the end of Step 2. Bringing the head down to lying in Step 2.1 and back up to upright in Step 2.3 is not this violation.
- **SOP rule broken:** Steps 2.1 and 2.3, never turn the pump head, and leave it seated square on the neck.
- **Coaching note:** straight up to remove, straight down from above to seat. The tilt onto its side and back is allowed; a spin about its axis never is, and the spout points wherever it lands.

Violation: Pump head laid down wrong

- **Visible cue:** the **right gripper** puts the head down anywhere but the **head spot**, on the caddy, on a towel, on the staging strip, leaning against the pump bottle or the measuring cup, or on bare table outside the fill area; drops it rather than lowering it until it rests; drags the dip tube along the tabletop instead of laying it down; lets the head roll or come to rest off the head spot and moves on; or lands the head with its dip tube standing off the tabletop so it is not lying stable.
- **SOP rule broken:** Step 2.1, the head is laid on its side on the head spot and nowhere else, in one smooth tilt, and rests still before the gripper opens.
- **Coaching note:** nothing holds the head up, so lay it flat on its own patch and let it settle. Lower the tube onto the table, do not sweep it across.

Violation: Soap spilled

- **Visible cue:** soap runs down the outside of the pump bottle, drips onto the fill spot, the head spot, the table, the caddy, or an item, or the **right gripper** tips the measuring cup away from the neck while it is still running.
- **SOP rule broken:** Step 2.2, pour with the pour lip over the neck and bring the measuring cup upright over the neck before moving it away.
- **Coaching note:** keep the lip over the neck from the first tip to the last drip.

Violation: Towel rolled the wrong way, folded, lifted off its lane, or left loose

- **Visible cue:** the **right gripper** takes the near edge, the far edge, or the left edge of the towel instead of its **right edge**; the turn is pushed away from the front table edge or toward it instead of from right to left; the finished roll's axis runs left to right instead of front to back; the towel is creased flat into a folded rectangle or laid on the shelf flat rather than wound into a roll; it is lifted off the tabletop and wound in the air, dragged out of its lane across the table, or rolled across the other towel's lane; or the finished roll springs open, hangs a loose flap, or unwinds on the shelf and the episode moves on.
- **SOP rule broken:** Step 3.2, the right gripper takes the right edge, the towel is rolled from right to left where it lies inside its own lane, and it is left tight so it holds its shape.
- **Coaching note:** right edge, and the turn runs across to the left so the finished roll points front to back. A loose roll opens on the shelf and a fat roll will not sit inside it.

Violation: Towel not held by the left gripper while it is rolled

- **Visible cue:** the **right gripper** rolls the towel while the **left gripper** waits clear of the table or hovers without taking hold; the left gripper holds somewhere other than the left edge, or holds the edge up off the tabletop instead of pinned flat; it lets go before the roll is near it, so the towel slides or bunches under the run; it stays down until the roll runs into it; it pushes the turn, rolls, or moves the towel; both grippers roll together; the left gripper comes in only part way through the roll; or the edge goes over skewed so the finished roll is crooked.
- **SOP rule broken:** Step 3.2, the left gripper holds the left edge flat while the right gripper rolls, and lifts clear only when the roll is nearly on it.
- **Coaching note:** one gripper holds, one gripper rolls. The left gripper pins the left edge still so the towel winds instead of sliding, and gets out of the way at the last moment.

Violation: Towel stack wrong

- **Visible cue:** the **right gripper** puts the second roll beside the first instead of on it, or leaves it perched on one end of the roll below, bridging the two, or resting partly on the roll and partly on the shelf floor or the raised divider; an end of the top roll hangs past the end of the roll below it; the rolls lie across each other rather than parallel; the top roll is pressed down hard enough to flatten either roll; a roll overhangs the caddy front or back, rests on or leans against the divider; or a roll ends up on the item shelf.

- **SOP rule broken:** Step 3.3 and **Stacked**, lower each roll straight down into the towel shelf and set the second one onto the first so it bears along its whole length with its ends level, both axes running front to back.
- **Coaching note:** both rolls run front to back, the top one landing along the whole length of the one below with its ends level, both fully inside the shelf and clear of the divider. A roll perched on one end works loose when the caddy is lifted.

Violation: Item out of the written order

- **Visible cue:** the **right gripper** stands an item in a place the written order gives to a different item, or two items end up swapped in the row. The order is shampoo, conditioner, deodorant, soap pump, left to right from the raised divider.
- **SOP rule broken:** Step 4, load left to right in the written order.
- **Coaching note:** nothing on the shelf tells you the order, so it comes from the SOP. The strip is scrambled on purpose, so pick by name rather than by whichever is nearest.

Violation: Row bunched or uneven

- **Visible cue:** the four items end up crowded into part of the shelf with a large empty run beside them, a gap between two items is visibly closed up while another is wide, an item sits so far over that the next one has no room, or the row runs into the raised divider or the right end wall.
- **SOP rule broken:** Step 4, the four items stand in one row across the shelf with a visible gap between each pair, the gaps roughly even by eye.
- **Coaching note:** the shelf is bare, so space the row against the divider and the end wall. Look at where the last item will go before placing the first.

Violation: Item not standing on its own

- **Visible cue:** an item leans, rocks, or is left tilted; its base sits over an edge; two items touch, an item rests against the raised divider or the end wall, or one only stays upright because a neighbour props it; or a toiletry stands with its label away from the front.
- **SOP rule broken:** Step 4, each item stands on its own base in its own place, clear of its neighbours and clear of the divider.
- **Coaching note:** every item stands on its own base with clear space beside it. Nothing on this shelf holds it up, and the divider is not a backstop.

Violation: Released before the item was standing

- **Visible cue:** the **right gripper** opens while the item is still tilted, still swinging, or not yet taking its own weight, and the item then rocks, leans, or topples.
- **SOP rule broken:** Step 4, hold the item a moment until it is clearly standing on its own, then open the gripper.

- **Coaching note:** let it settle on its base first. On a bare shelf the release is the moment the placement is won or lost.

Violation: Item turned in the air

- **Visible cue:** an item is lifted and rotated in the gripper to change the way it faces, or the pump head or the pump bottle is spun mid carry. Laying the pump head onto its side in Step 2.1 and bringing it upright in Step 2.3 is not this violation: that is a tilt, not a change of facing.
- **SOP rule broken:** Steps 2 and 4, never turn an item in the air.
- **Coaching note:** the gripper cannot reorient an item in the air. Stand it down, let go, and take it again with the facing wanted. The head's tilt onto its own patch is the one attitude change written in.

Violation: Item gripped by its cap, lid, or head

- **Visible cue:** the **right gripper** closes on a toiletry cap or lid, or on the pump head, to lift the item it belongs to.
- **SOP rule broken:** Steps 4.2 and 4.3, never grip a toiletry by its cap or lid and never lift the pump bottle by its head.
- **Coaching note:** grip the body. An item carried by its cap can come apart mid carry.

Violation: Bad pick, carry, or release

- **Visible cue:** an item is pushed, slid, or scraped across the table, the staging strip, a towel lane, or a shelf instead of being lifted clear and carried; two or more items are carried in one trip; the measuring cup is set down leaning, on its side, or off its cup spot, or is left in the **right gripper** while another action starts; a gripper opens over table outside a release point, over another item, over the raised divider, or over a place the held item does not belong in; or an item is dropped from above rather than set down.
- **SOP rule broken:** Steps 2 to 4, one item per trip, lifted clear of the surface and carried, and released only over the spot it is going to.
- **Coaching note:** the item leaves the surface before it moves, and where it lands is known before the gripper opens.

Violation: Placement shoved or fiddled with

- **Visible cue:** a gripper nudges, pats, or slides an item, a roll, or the pump head into position after releasing it instead of lifting it clear and standing it down again; a gripper closes on an item or the cloth, finds the grip off, and closes again more than twice before lifting; or more than about two small corrections are made to settle a placement. The recoveries written in Steps 3 and 4 do not count.
- **SOP rule broken:** Steps 1 to 4, correct a bad placement by lifting the item clear and standing it down again, in one or two small corrections.

- **Coaching note:** lift it and stand it again rather than shoving it. If the first placement lands wide every time, the approach needs tightening rather than correcting after.

Violation: Required check not followed

- **Visible cue:** a failed check is visible and the episode continues without the written correction, or a recovery the SOP does not write is used.
- **SOP rule broken:** Steps 1 to 4, perform each check and use only its stated correction.
- **Coaching note:** stop at each check and use only its written recovery.

Violation: Dropped or knocked over

- **Visible cue:** an item, a towel, or the cloth falls; a toiletry topples on the staging strip or the shelf; the **right gripper** knocks over a toiletry still standing on the strip while picking a neighbour; the pump bottle or the measuring cup is knocked over, shifted off its spot, or tipped while the soap is poured; a placed item is knocked over on approach or retract; a roll going in or coming out unseats the roll already inside; or an item is pushed off a shelf, over the raised divider, or off the table.
- **SOP rule broken:** Steps 1 to 4, nothing is dropped, toppled, or knocked out of its place, and an item is released only after a settled placement.
- **Coaching note:** check the path and the landing spot before the arm moves, and retract straight up out of the shelf so the gripper does not catch anything.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with dirt on either shelf, the caddy left at its back place or off square, fewer than four items on the item shelf, an item left on the staging strip or in the towel area, two items touching, the row bunched at one end, soap left in the measuring cup, the pump head unseated, the cloth left on the cloth spot or anywhere off its park, an arm short of home, or a gripper not fully open.
- **SOP rule broken:** Step 5, confirm the caddy, the soap, and the zones, then return both arms home with grippers open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do before the recording ends.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Camera cannot see into a shelf from above**, so what is on that shelf or which side of the divider a thing landed on cannot be judged.

- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective toiletry:** one that will not stand upright unaided on a bare flat surface, slips out of a correct grip, or has a cap that will not stay tight.
- **Defective soap pump:** a head that will not seat under a correct straight press or binds under a correct straight lift, a head that rolls off its own patch or will not lie still on its side on bare tabletop, or a bottle that will not stand upright unaided holding its soap plus the cup's pour.
- **Defective measuring cup:** a pour lip that will not pour cleanly, runs so fast it cannot be poured without spilling, or drips after a correct upright return; a handle a correct grip will not hold; or a cup that will not stand upright unaided on bare tabletop full or empty.
- **Defective towel:** one so stiff, slick, or worn that it will not hold a roll after a correct held roll and press.
- **Defective cloth:** one that is torn, shedding, dry at episode start, or so loaded with dirt that a correct wipe only smears it.
- **Caddy will not lift level:** it hangs off level, swings, or twists on a correct grip at the middle of its raised divider, is too heavy for one arm to lift and carry loaded, or spills a correctly stacked roll out of the towel shelf under a correct level carry. Or it slides so freely on the tabletop that a correct wipe or a correct straight down placement moves it off its place.
- **Raised divider loose, weak, or too tall:** a divider that shifts under a correct wipe, that flexes, works loose, or parts from the caddy under a correct lift, that is too thick or too thin for the gripper to close on, or that stands so tall a correct straight down approach cannot clear it or the camera cannot read past it.
- **Shelf will not hold an item:** a shelf that is not level, or so slick or so shallow front to back that a correctly stood item or a correctly placed roll will not stay where it is put.
- **Towel shelf too tight:** a shelf so narrow or so short that a correctly rolled towel will not sit inside it clear of the divider, or so shallow that the second roll will not stay on the first after a correct straight down placement.
- **Staging strip will not hold a toiletry:** tabletop that is not level, or so slick that a correctly stood toiletry will not stay standing on the bare surface.
- **Fill area will not hold the bottle, the cup, or the head:** tabletop that is not level, or so slick that a correctly stood pump bottle or measuring cup slides off its spot, that the bottle tips or walks under a correct pour with nothing touching it, or that a correctly laid pump head rolls away from the head spot.
- **Towel area will not hold a towel:** tabletop so slick that a towel slides along the table instead of winding up under a correct roll with the left gripper holding, or that will not hold a flat towel square in its lane.
- **Damaged shelf surface:** a stain, scratch, or burn in the shelf itself that does not lift under a correct wipe.

- **A zone shifts during the episode**, or a zone turns out to sit outside its arm's comfortable reach so a shelf at either of the caddy's two places, a place in the row, or a place in the fill area cannot be reached without extending or folding the arm.

Annotation subtasks (from SOP)

1. Take the cloth from the cloth park
2. Wipe the towel shelf and set the cloth on the cloth spot
3. Take the cloth from the cloth spot and wipe the item shelf
4. Return the cloth to the cloth park
5. Lay the pump head on the head spot
6. Pour soap from the measuring cup into the soap pump bottle
7. Seat the pump head in the soap pump bottle
8. Lift the caddy to its back place
9. Roll one hand towel
10. Lower one rolled towel into the towel shelf
11. Lift the caddy to its home place
12. Place one toiletry in its position
13. Place the soap pump bottle in its position
14. Return both arms home and end the episode